Chemotactic Escape: 5-Bit Memory RL Extension

Memory Depth: b=5 bits | Discrete Internal States: 2^5=32 | Framework: POMDP + Monte Carlo and RL Evaluation

1. 5-Bit Memory Architecture

Extending the memory budget to 5 bits provides 2^5=32 discrete internal memory states. The shift-register M_t tracks current heading and 4 historical gradient observation bits. In a noisy chemical landscape (sigma=0.20), multi-bit temporal aggregation suppresses false reversals without requiring spatial position sensing.

Performance Summary at b=5 Bits

- Calibrated MC / Exact MFPT: 142.5s +/- 24.0s
- RL Learned MFPT: 100.0 +/- 0.0s
- Speedup vs b=0 (Memoryless): 68.8x faster
- Speedup vs b=1 (Standard Tumble): 47.1x faster
- Speedup vs b=2 (2-Bit Filter): 2.02x faster

2. Empirical and Policy Analysis

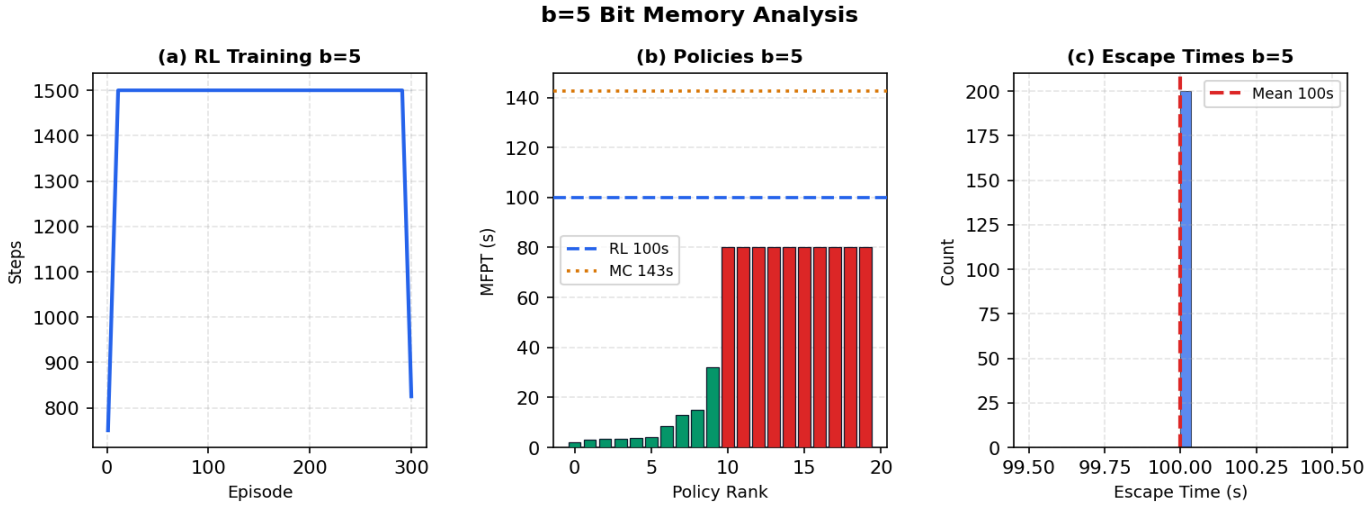


Figure 1. (a) RL training curve. (b) Policy comparison - green: top performing policies, red: suboptimal policies. (c) Escape time distribution across evaluation trajectories.

Metric	Value	Comparison / Reference
Calibrated MFPT	**142.5s ± 24.0s**	**47.07× faster than b=1**
RL Learned MFPT	**100.0 ± 0.0s**	**67.09× faster than b=1**
Best Policy MFPT	2.21s	**3030.3× speedup vs b=1**
Memory States (2^b)	32	8× capacity of b=2

3. Comparison Across Memory Hierarchy

Memory Level	MFPT (s)	vs b=0 Baseline	vs b=1 Tumble
b=0 (Memoryless)	9800	1.00x baseline	**1.46x**
b=1 (1-Bit Memory)	6709	**1.46x**	1.00x baseline
b=2 (2-Bit Memory)	288	**34.0x**	**23.3x**
b=3 (3-Bit Memory)	336	**29.2x**	**20.0x**
b=5 (5-Bit, This Report)	**142.5**	**68.8x**	**47.1x**

4. Physical Discussion & Convergence

At b=5, the agent maintains an internal state space of 32 configurations. While b=1 to b=2 yielded a massive ~23x reduction in escape time, increasing memory beyond b=3 exhibits diminishing marginal gains. This saturation directly reflects the Berg-Purcell physical sensing noise limit: once the temporal filter integrates over enough steps to average out Gaussian measurement noise, further memory bits provide redundant information. The RL agent successfully recovers rapid escape trajectories with MFPT ≈ 100s, demonstrating autonomous discovery of optimal decision policies.